

**8. Department of Science and Technology – IX
Regional Standards and Testing Laboratory**

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Field of Testing : Chemical Testing and
Microbiological Testing

Chemical Testing

Products	Specific Test	Method/Reference Standard
1. Foods		
.01 Cereal and Cereal Products		
1. Breakfast Cereals 2. Cereal/Cereal Grains 5. Flour, corn meal, corn grits, semolina 10. Starch	Moisture	AOAC 925.10
	Ash	AOAC 923.03
	Carbohydrates	AOAC 986.25 E
	Energy	Atwater Factor
	Crude Protein	AOAC 2001.11
	Sodium	AOAC 985.35
	Total Fat	AOAC 2003.05
.02 Nuts and Nut products		
Peanut Butter and other nut Butters	Moisture	AOAC 925.40
	Ash	AOAC 942.05
	Carbohydrates	AOAC 986.25 E
	Energy	Atwater Factor
	Crude Protein	AOAC 2001.11
	Sodium	AOAC 985.35
	Total Fat	AOAC 2003.05
.03 Dairy Products		
6. Milk Powders 7. Sweetened Condensed Milk 8. Liquid Milk 9. Pasteurized Milk	Carbohydrates	AOAC 986.25 E
	Energy	Atwater Factor
	Crude Protein	AOAC 2001.11
	Water Activity	AOAC 978.18

	Sodium	AOAC 985.35
	Moisture	AOAC 927.05 (powder) AOAC 990.19 (liquid)
	Ash	AOAC 930.30 (powder) AOAC 945.46 (liquid)
	Total Fat	AOAC 2003.05
.04 Meat and Meat Products		
1. Dried Animal Products	Moisture	AOAC 950.46 B
2. Meat Paste and Pate (heat treated)	Ash	AOAC 920.153
3. Cold Cuts, frozen and chilled hotdogs, corned beef, luncheon meat	Carbohydrates	AOAC 986.25 E
	Energy	Atwater Factor
4. Packaged cooked cured/salted meat (ham, bacon)	Crude Protein	AOAC 2001.11
5. Fermented, comminuted meat, not cooked (dry and semi-dry fermented sausages)	Sodium	AOAC 985.35
6. Cooked poultry meat, frozen to be re-heated before eating	Total Fat	AOAC 2003.05
7. Cooked poultry meat, frozen, ready to eat		
8. Cured/smoked poultry meat		
9. Dehydrated Poultry Products		
10. Fresh/Frozen raw chicken		
11. Meat Products in hermetically sealed containers		
.05 Fish and Fish products, including mollusks, crustaceans, and echinoderms		
1. Fresh and frozen fish and cold-smoked	Moisture	AOAC 952.08
2. Frozen cooked crustaceans	Ash	AOAC 938.08
5. Fish and shellfish products in hermetically sealed containers	Carbohydrates	AOAC 986.25 E
6. Smoked, dried, fermented, and/or salted fish and fish products	Energy	Atwater Factor
	Crude Protein	AOAC 2001.11
	Sodium	AOAC 985.35
	Lead	AOAC 999.11
	Cadmium	AOAC 999.11
	Total Fat	AOAC 2003.05
.07 Confectioneries		
1. Cocoa Powder	Moisture	AOAC 931.04

2. Chocolate Products Chocolate 3. Confectioneries (chocolate bars, blocks, bonbons)	Ash	AOAC 972.15
	Carbohydrates	AOAC 986.25 E
	Energy	Atwater Factor
	Crude Protein	AOAC 2001.11
	Sodium	AOAC 985.35
	Lead	AOAC 999.11
	Cadmium	AOAC 999.11
	Total Fat	AOAC 2003.05
.08 Fruits, jams and other fruit products		
1. Sun dried fruit 2. Jam, Jellies, marmalades 3. Fruit-based spreads 4. Candied Fruit 5. Fruit preparations 6. Fermented Fruit Products	Moisture	AOAC 920.151 A (fresh and canned, jams, marmalades, and preserves) AOAC 925.45 D (fruit juices, jellies, and syrups)
	Ash	AOAC 940.26
	Carbohydrates	AOAC 986.25 E
	Energy	Atwater Factor
	Crude Protein	AOAC 2001.11
	Sodium	AOAC 985.35
	Lead	AOAC 999.11
	Total Fat	AOAC 2003.05
.09 Vegetables and vegetable products		
1. Dried Vegetables 4. Canned or bottled (pasteurized) or retort pouch vegetables 6. Cooked or fried vegetables	Moisture	AOAC 964.22
	Ash	AOAC 925.51
	Carbohydrates	AOAC 986.25 E
	Energy	Atwater Factor
	Crude Protein	AOAC 2001.11
	Sodium	AOAC 985.35
	Lead	AOAC 999.11
	Cadmium	AOAC 999.11
	Total Fat	AOAC 2003.05
.12 Fruit juices, drinks and Concentrates		
4. Coffee, coffee substitutes, tea, herbal infusions, and other hot	Lead	AOAC 999.11

cereal and grain beverages, excluding cocoa	Cadmium	AOAC 999.11
.15 Eggs and egg products		
1. Egg products 2. Egg Based desserts	Moisture	AOAC 925.30
	Ash	AOAC 920.153
	Carbohydrates	AOAC 986.25 E
	Energy	Atwater Factor
	Crude Protein	AOAC 2001.11
	Sodium	AOAC 985.35
	Total Fat	AOAC 2003.05
.16 Bakery Products		
1. Frozen bakery Products (ready to eat) 2. Frozen bakery Products (to be cooked) 5. Baked Goods	Moisture	AOAC 925.10
	Ash	AOAC 923.03
	Carbohydrates	AOAC 986.25 E
	Energy	Atwater Factor
	Crude Protein	AOAC 2001.11
	Sodium	AOAC 985.35
	Total Fat	AOAC 2003.05
.20 Other Food Products		
Fried Snack Foods	Moisture	AOAC 925.10
	Total Fat	AOAC 2003.05
Philippine Tablea	pH	AOAC 970.21
	Moisture	AOAC 931.04
	Crude Fat	AOAC 2003.05
0.24 Sauce, Spices and condiments		
1. Spice and Herbs	Moisture	AOAC 120.171
	Lead	AOAC 999.11
	Cadmium	AOAC 999.11
5. Vinegars	pH	AOAC 960.19
7. Sauces and like products (e.g.fish sauce)	pH	AOAC 960.19
	Crude Protein	AOAC 2001.11
8. Soybean-based seasonings and condiments (e.g. soy sauce)	Total Nitrogen	AOAC 2001.11

4. Water		
.01 Bottled Water	pH	SMEWW 4500-H ⁺ B
	Lead	SMEWW 3111 B
	Cadmium	SMEWW 3111 B
	Copper	SMEWW 3111 B
	Iron	SMEWW 3111 B
	Total Dissolved Solids	SMEWW 2540 C
	Zinc	SMEWW 3111 B
	Calcium	SMEWW 3111 B
	Magnesium	SMEWW 3111 B
	Sodium	SMEWW 3111 B
	Potassium	SMEWW 3111 B
	Chloride	SMEWW 4500-Cl ⁻ B
	Sulfate	SMEWW4500-SO ₄ ⁻² E

Legend to Reference/Methods:

AOAC – Association of Official Analytical Chemists

SMEWW- Standard Methods for the Examination of Water and Waste Water

*** *Nothing follows* ***

Microbiological Testing

Products	Specific Test	Method/Reference Standard
1. FOODS		
1. MILK AND DAIRY PRODUCT		
01.1 Milk Powders (e.g. whole, nonfat or filled milk, buttermilk, whey & whey protein concentrate) intended for children more the 36 months of age and adults)	<i>Salmonella</i>	Reveal 2.0 for Salmonella
01.2 Sweetened Condensed Milk/Creamer	Coliforms Count, cfu/g	Pour Plate Method/ BAM Online Chapter 4
	Yeast & Mold Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
01.3 Liquid Milk (evaporated or ready to drink) & Cream (UHT/sterilized)	Commercial sterility	BAM Online Chapter 21A
01.4 Pasteurized Milk	Coliforms Count, cfu/mL	Pour Plate Method/ BAM Online Chapter 4
	<i>Salmonella</i>	Reveal 2.0 for Salmonella
	Aerobic Plate Count, cfu/mL	Pour Plate Method/ BAM Online, Chapter 3
01.5 Pasteurized Cream	Coliforms Count, cfu/g	Pour Plate Method/ BAM Online Chapter 4
	<i>Salmonella</i>	Reveal 2.0 for Salmonella
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
01.6 Yogurt and other fermented milk	<i>S. aureus</i> , cfu/mL	Spread Plate Technique/BAM Online
	<i>Salmonella</i>	Reveal 2.0 for Salmonella
	Coliforms Count, cfu/mL	Pour Plate Method/ BAM Online Chapter 4
01.7 Heat treated fermented milks (sterilized, UHT)	Commercial sterility	BAM online Chapter 21A
01.8 Processed Cheese Spread	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	Coliforms Count, cfu/g	Pour Plate Method/ BAM Online Chapter 4
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
01.9 Soft cheese (from pasteurized milk)	<i>E. coli</i> , cfu/g	Conventional Method/ BAM Online Chapter 4
	<i>Salmonella</i>	Reveal 2.0 for Salmonella
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
01.10 Hard and semi-hard cheese	<i>E. coli</i> , cfu/g	Conventional Method/ BAM Online Chapter 4

Products	Specific Test	Method/Reference Standard
	<i>Salmonella</i>	Reveal 2.0 for Salmonella
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
01.11 All Raw Milk cheese; raw milk un-ripened cheese w/ moisture >50%, pH >5.0	<i>Salmonella</i>	Reveal 2.0 for Salmonella
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
01.12 Cream Cheese products	Aerobic Plate Count	Pour Plate Method/ BAM Online, Chapter 3
	Coliforms, cfu/g/MPN/g	Pour Plate Method/ MPN Method/ BAM Online Chapter 4
	<i>E. coli</i> , cfu/g/MPN/g	Conventional/ MPN Method/ BAM Online Chapter 4
	Yeast & Mold Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
01.13 Cream cheese products	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	Coliforms, cfu/g/MPN	Pour Plate Method/ MPN Method/ BAM Online Chapter 4
	<i>E. coli</i> , cfu/g/MPN	Pour Plate Method/ MPN Method/ BAM Online Chapter 4
2. FATS, OILS AND AND FAT EMULSIONS		
02.1 Butter (whipped, pasteurized)	Yeast & Mold Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Coliforms, cfu/g	Pour Plate Method/ BAM Online Chapter 4
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
02.2 Butter made from unpasteurized milk or unpasteurized milk products	Coliforms, cfu/g	Pour Plate Method/ BAM Online Chapter 4
	<i>E. coli</i> , MPN/g	MPN Method/ BAM Online Chapter 4
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	<i>Salmonella</i>	Reveal 2 for Salmonella
	Aerobic Plate Count	Pour Plate Method/ BAM Online, Chapter 3
02.3 Margarine	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	<i>Salmonella</i>	Reveal 2 for Salmonella
	Aerobic Plate Count	Pour Plate Method/

Products	Specific Test	Method/Reference Standard
		BAM Online, Chapter 3
	Yeast & Mold Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
02.5 Virgen coconut oil	Aerobic Plate Count, cfu/ml	Pour Plate Method/ BAM Online, Chapter 3
	Coliform, cfu/mL, MPN/mL	Pour Plate /MPN Method BAM Online Chapter 4
	Yeast & Mold Count, cfu/mL	Spread Plate Technique/ BAM Online Chapter 18
	<i>Salmonella</i>	Reveal 2 for Salmonella
	<i>E. coli</i> , MPN/ml/cfu/mL	Pour Plate /MPN Method BAM Online Chapter 4
3. EDIBLE ICES, INCLUDING SHERBET AND SORBET		
03.1 Ice cream & Sherbet (plain and flavored)	Coliform, cfu/g	Pour Plate Method/ BAM Online Chapter 4
	<i>Salmonella</i>	Reveal 2 for Salmonella
03.2 2 Ice cream with added ingredients (nuts, fruits, cocoa, etc.	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
03.3 Flavored Ice (e.g. ice candy)	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	Coliform, Cfu/mL, MPN/mL	Pour Plate /MPN Method BAM Online Chapter 4
	Yeast & Mold Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	<i>Salmonella</i>	Reveal 2 for Salmonella
4. CONFECTIONARIES		
04.1 Cocoa powder	Mold, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
04.2 Chocolate Confectionaries (chocolate bars, blocks, bonbons)	<i>Salmonella</i>	Reveal 2 for Salmonella
	Coliform, cfu/g, MPN/g	Pour Plate /MPN Method BAM Online Chapter 4
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
04.3 Sugar Confectionaries (Hard and soft candies, toffees, caramel, fondants, creams, nougats and paste)	Molds, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Coliform, cfu/g, MPN/g	Pour Plate /MPN Method BAM Online Chapter 4
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
04.4 Ethnic Milk-based Confectionaries (Pastillas and Yema)	Mold & Yeast Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	<i>Salmonella</i>	Reveal 2 for Salmonella
	Coliform, cfu/g, MPN/g	Pour Plate /MPN Method

Products	Specific Test	Method/Reference Standard
		BAM Online Chapter 4
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
04.5 Ethnic Flour-based Confectionaries (Polvoron, Piaya and Barquillos)	Mold &Yeast Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Coliform, cfu/g	Pour Plate Method/ BAM Online Chapter 4
04.6 Coconut sap sugar	<i>Salmonella</i>	Reveal 2 for Salmonella
	Coliform, cfu/g, MPN/g	Pour Plate /MPN Method BAM Online Chapter 4
	E. coli, cfu/g/MPN/g	Pour Plate Method/ MPN Method/ BAM Online Chapter 4
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
04.7 Dry, granulated sugar 04.8 Liquid sugar (sugar syrup)	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	Yeast Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Mold, cfu/g	
04.8 Non-nutritive sweeteners (e.g aspartame, sucralose and bulk sweeteners (e.g. polysols)	Coliform, cfu/g, MPN/g	Pour Plate /MPN Method BAM Online Chapter 4
	E. coli, cfu/g/MPN/g	Pour Plate Method/ MPN Method/ BAM Online Chapter 4
	<i>Salmonella</i>	Reveal 2 for Salmonella
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
5.0 FRUITS AND VEGETABLES, NUTS AND SEEDS		
05.1 Frozen Vegetables & fruits (pH >4.5)	E. coli, cfu/g	Conventional method BAM Online Chapter 4
05.2 Fermented Vegetable Ready to Eat (e.g. Kimchi)	Mold &Yeast Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Coliform, cfu/g, MPN/g	Pour Plate/ MPN Method BAM Online Chapter 4
	<i>E. coli</i> , cfu/g/MPN/g	Pour Plate Method/ MPN Method/ BAM Online Chapter 4
	<i>Salmonella</i>	Reveal 2 for Salmonella
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
05.3 Fruits & Vegetable products in hermetically sealed containers (thermally processed)	Commercial Sterility	BAM Online Chapter 21A
05.4 Dried Vegetables	<i>E. coli</i> , cfu/g	Pour Plate Method/ BAM Online Chapter 4
05.5 Coconut	<i>Salmonella</i>	Reveal 2 for Salmonella

Products	Specific Test	Method/Reference Standard
(desiccated)	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	Coliform, cfu/g	Pour Plate Method/ BAM Online Chapter 4
	Mold & Yeast Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	<i>E. coli</i> , cfu/g/MPN/g	Pour Plate Method/ MPN Method/ BAM Online Chapter 4
05.6 Peanut Butter & other Nut Butters	<i>Salmonella</i>	Reveal 2 for Salmonella
05.7 Sun Dried Fruits	Mold, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	<i>E. coli</i> /MPN/g	MPN Method/ BAM Online Chapter 4
05.8 Dried Fruits (e/g dried mangoes, dates, figs, apricots, raisins, etc.)	Yeast & Molds, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
05.9 Ethnic food products sweet preserves in syrup (e.Kaong, Langka, Banana, Legumes, chickpeas, red beans, white kidney beans, coconut, nata de coco, mixed preserves (halo-halo))	Acidified sweet preserves: $\text{pH} \leq (a_w)$	
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	Yeast & Molds Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Low acid sweet preserves: $\text{pH} > 4.6; (a_w) > 0.85$	
	Commercial Sterility	BAM Online Chapter 21A
05.10 Purple Yam (Ube) Jam (Halaya)	Commercial Sterility	BAM Online Chapter 21A
05.11 Malunggay (Moringa) powder	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	Yeast & Molds, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	<i>E. coli</i> /MPN/g	MPN Method/ BAM Online Chapter 4
	<i>Salmonella</i>	Reveal 2 for Salmonella
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
6 EGG AND EGG PRODUCTS		
6.1 Pasteurized egg	Coliforms, cfu/g	Pour Plate Method/ BAM Online Chapter 4
	<i>Salmonella</i>	Reveal 2 for Salmonella
	Yeast & Molds Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3

Products	Specific Test	Method/Reference Standard
7. CEREALS AND CEREAL PRODUCTS		
07.1 Breakfast cereals	Molds, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Yeast & Yeast-like fungi, cfu/g	
	Coliforms cfu/g	Pour Plate Method/ BAM Online Chapter 4
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
07.2 Cereals/Cereal Grains	Yeast & Molds Count, cfu/g	Spread Plate Technique/BAM Online
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	Coliforms, cfu/g	Pour Plate Method/ BAM Online Chapter 4
	E. coli, cfu/g	Conventional Method BAM Online Chapter 4
07.5 Flour, corn meal, corn grits, Semolina	Molds, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Yeast & Yeast-like fungi, cfu/g	
	Coliforms, cfu/g	Pour Plate Method/ BAM Online Chapter 4
07.6 Dry mixes (e.g. Pancake mix, Purple, yam mix)	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	Yeast & Molds Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Coliforms, cfu/g	Pour Plate Method/ BAM Online Chapter 4
07.7 Soy protein	Coliforms, cfu/g	Pour Plate Method/ BAM Online Chapter 4
	<i>E. coli</i> , cfu/g	
	Yeast & Molds Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	<i>Salmonella</i>	Reveal 2 for Salmonella
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
07.8 Tufo	<i>E. coli</i> , cfu/g/MPN/g	Conventional MPN Method/ BAM Online Chapter 4
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
07.9 Pasta Products & Noodles Uncooked (wet and dry)	Coliforms cfu/g	Pour Plate Method/ BAM Online Chapter 4
	Yeast & Molds Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	Aerobic Plate Count, cfu/g	Pour Plate Method/

Products	Specific Test	Method/Reference Standard
		BAM Online, Chapter 3
07.10 Starch	Coliforms cfu/g	Pour Plate Method/ BAM Online Chapter 4
	Yeast & Molds Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	<i>Salmonella</i>	Reveal 2 for Salmonella
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
07.11 Instant noodles (noodle cake) 07.12 Instant Noodles (mixed with seasoning)	Coliforms cfu/g	Pour Plate Method/ BAM Online Chapter 4
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	<i>Salmonella</i>	Reveal 2 for Salmonella
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	Yeast & Molds Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
07.13 Rice based desserts (rice cakes), ready to eat e.g. kakanin, suman, layered, moulded rice balls, mochi	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	<i>E. coli</i> , cfu/g/MPN/g	Conventional MPN Method/ BAM Online Chapter 4
	<i>Salmonella</i>	Reveal 2 for Salmonella
8. BAKERY PRODUCTS		
08.1 Frozen bakery	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	<i>Salmonella</i>	Reveal 2 for Salmonella
08.2 Frozen & refrigerated doughs	<i>Salmonella</i>	Reveal 2 for Salmonella
08.3 Baked goods (e.g breads, cakes, pie crust, pastries and fried doughs)	Yeast Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Mold Count, cfu/g	
	Coliforms cfu/g	Pour Plate Method/ BAM Online Chapter 4
	<i>Salmonella</i>	Reveal 2 for Salmonella
08.4 Coated or filled, dried shelf- stable biscuits	Coliforms cfu/g	Pour Plate Method/ BAM Online Chapter 4
	Salmonella Detection	Reveal 2 for Salmonella
9. READY TO EAT SAVOURIES		
09.1 Snack foods (e.g. potato chips, popcorn, pretzels, rice crackers, flavored crackers, processed whole nuts, yoghurt-cereal and honey-covered nuts, dried fruits and cereal	Mold Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Yeast & Yeast-like fungi, cfu/g	
	Coliforms cfu/g	Pour Plate Method/ BAM Online Chapter 4
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3

Products	Specific Test	Method/Reference Standard
snacks, crackers with fish, fish product or flavoring)		
10. MEAT AND MEAT PRODUCTS		
10.1 Dried Animal Products	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	<i>Salmonella</i>	Reveal 2 for Salmonella
10.2 Cold cuts, frozen & chilled hot dogs, corned beef, luncheon meat	<i>E. coli</i> , cfu/g/MPN/g	Conventional MPN Method/ BAM Online Chapter 44
	<i>Salmonella</i>	Reveal 2 for Salmonella
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	Coliforms cfu/g	Pour Plate Method/ BAM Online Chapter 4
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
10.3 Packaged cooked cured/salted meat (ham, bacon)	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	<i>Salmonella</i>	Reveal 2 for Salmonella
10.4 Fermented, comminuted meat, not cooked (dry & semi-dry fermented sausages)	<i>E. coli</i> /MPN/g	Conventional MPN Method/ BAM Online Chapter 4
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	<i>Salmonella</i>	Reveal 2 for Salmonella
10.5 Cooked poultry meat, frozen to be reheated before eating (e. prepared frozen meals chicken burgers, chicken turkey rolls, chicken nuggets, other breaded poultry meat products	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	<i>Salmonella</i>	Reveal 2 for Salmonella
10.6 Cured/smoked poultry meat	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	<i>Salmonella</i>	Reveal 2 for Salmonella
10.7 dehydrated poultry products	<i>Salmonella</i>	
10.8 Meat products in hermetically sealed products	Commercial sterility	BAM Online Chapter 21A
10.9 Marinated meat products (e.g. Marinated meat and meat preparations (tapa, sisig, etc.) Marinated poultry. Dimsum made from meat (somai)	<i>Salmonella</i>	Reveal 2 for Salmonella
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
10.9 Minced meat & meat preparation made from poultry meat intended to be eaten cooked	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online Chapter 3
	<i>E. coli</i> , cfu/g	Conventional BAM Online Chapter 4
	<i>Salmonella</i>	Reveal 2 for Salmonella

Products	Specific Test	Method/Reference Standard
10.10 Minced meat and meat preparation made from spices other poultry intended to be eaten	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online Chapter3
	<i>E. coli</i> , cfu/g	Conventional BAM Online Chapter 4
	<i>Salmonella</i>	Reveal 2 for Salmonella
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
10.11 Foods cooked immediately prior to sale or consumption (takeaway foods, burgers, kebab, sausages, pizza, ready meals (cook/chill and cook/freeze) after regeneration	Aerobic Plate Count, cfu/g	Pour Plate Method BAM Online Chapter 3
	<i>E. coli</i> , cfu/g	Conventional BAM Online Chapter 4
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	<i>Salmonella</i>	Reveal 2 for Salmonella
11. FSIH AND OTHER SEAFOOD PRODUCTS		
11. 1 Fresh frozen fish & cooked -smoked 11.3 Frozen raw crustaceans 11.4 Frozen cooked crustacean	<i>E. coli</i> , MPN/g	MPN Method/ BAM Online Chapter 4
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	<i>Salmonella</i>	Reveal 2 for Salmonella
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
11.5 Pre-cooked breaded fish 11.6 Cooked, chilled & frozen crabmeat	<i>E. coli</i> , MPN/g	MPN Method/ BAM Online Chapter 4
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
11.7 Cooked crustaceans	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	<i>Salmonella</i>	Reveal 2 for Salmonella
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
11.8 Fresh & frozen Bivalve molluscs	<i>E. coli</i> , MPN/g	MPN Method/ BAM Online Chapter 4
	<i>Salmonella</i>	Reveal 2 for Salmonella
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
11.9 Fish & shellfish products, cooked crustaceans in hermetically sealed containers (thermally processed e.g. cooked bagoong/shrimp paste	Commercial sterility	BAM Online Chapter 21A
11.10 Ethnic foods products, dried, salted fish	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	Yeast & Mold Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18

Products	Specific Test	Method/Reference Standard
	Coliforms, MPN/g	MPN Method/ BAM Online Chapter 4
	<i>E. coli</i> , MPN/g	
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
11.11 Smoked fish	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	<i>Salmonella</i>	Reveal 2 for Salmonella
	<i>E. coli</i> , MPN/g	MPN Method/ BAM Online Chapter 4
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
11.12 Salt fermented fish & shrimp (bagoong)	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	Coliforms. cfu/g	
11.13 Thermally processed fish products	Commercial sterility	BAM Online Chapter 21A
11.14 Aquatic products	<i>Salmonella</i>	Reveal 2 for Salmonella
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
11.15 Fish & crustacean based processed meat (fish ball, squid ball)	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	<i>E. coli</i> , MPN/g	MPN Method/ BAM Online Chapter 4
12. SPICES, SOUPS, SAUCES, SALAD AND PROTEIN PRODUCTS		
12.1 Dry mixes for Soup and sauces	Yeast & Mold Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	Coliforms. cfu/g	
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	<i>Salmonella</i>	Reveal 2 for Salmonella
12.2 Yeast	<i>Salmonella</i>	Reveal 2 for Salmonella
12. 3 Spices	Mold, cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
12.4 Spices (ready to eat)	Coliforms. cfu/g	Pour Plate Method/ BAM Online Chapter 4
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	<i>Salmonella</i>	Reveal 2 for Salmonella
	Mold, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
12.5 Food grade gelatin	<i>E. coli</i> , cfu/g/MPN/g	Pour Plate Method

Products	Specific Test	Method/Reference Standard
12.6 Emulsified sauce		MPN Method BAM Online Chapter 4
	<i>Salmonella</i>	Reveal 2 for Salmonella
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
12.7 Instant noodles pH < 4.6 e.g. Mayonnaise, Thousand islands, Ranch, French)	Yeast & Mold Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	<i>Salmonella</i>	Reveal 2 for Salmonella
	Coliforms. cfu/g	Pour Plate Method/ BAM Online Chapter 4
12.8 Instant noodles seasoning	Yeast & Mold Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	<i>Salmonella</i>	Reveal 2 for Salmonella
12.9 Ready to eat flavorings/Instant condiments (e.g. soy sauce and sauce and sauce products, composite seasonings)	Coliforms. cfu/g	Pour Plate Method/ BAM Online Chapter 4
	Yeast & Mold Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
12.10 Salads and sandwich spreads (excluding cocoa milk- based sandwich spreads)	<i>Salmonella</i>	Reveal 2 for Salmonella
	<i>S. aureus</i> , cfu/g	Spread Plate Technique/ BAM Online Chapter 12
	<i>Salmonella</i>	Reveal 2 for Salmonella
13. BEVERAGES		
13.1 Non alcoholic beverages (e.g. ready to drink, softdrinks, ices tea, energy drinks, jelly drinks)	Yeast & Mold Count, cfu/mL	Spread Plate Technique/ BAM Online Chapter 18
	Coliforms, cfu/mL	Pour Plate Method/ BAM Online Chapter 4
	Aerobic Plate Count, cfu/mL	Pour Plate Method/ BAM Online, Chapter 3
13.2 Frozen juice concentrate	Aerobic Plate Count, cfu/gmL	Pour Plate Method/ BAM Online, Chapter 3
	Yeast & Mold Count, cfu/mL	Spread Plate Technique/ BAM Online Chapter 18
13.3 Juices in hermetically sealed containers (tetra pack, etc)	Commercial sterility	BAM Online Chapter 21A
13.4 Powdered beverages	Aerobic Plate Count, cfu/g	Pour Plate Method/

Products	Specific Test	Method/Reference Standard
(e.g. iced tea, powdered juices/mixes)		BAM Online, Chapter 3
	Yeast & Mold Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	Coliforms. cfu/g	Pour Plate Method/ BAM Online Chapter 4
13.5 Fruits beverages products	Aerobic Plate Count, cfu/g	Pour Plate Method/ BAM Online, Chapter 3
	Yeast & Mold Count, cfu/g	Spread Plate Technique/ BAM Online Chapter 18
	<i>E. coli</i> , cfu/mL	Conventional BAM Online Chapter 4
13.6 Bottled water (Purified/Distilled water)	Heterotrophic Plate Count cfu/mL	Pour Plate Method 9215B, SMEWW 2017 23 rd ed
	Coliforms, MPN/100 mL	Multiple Tube Fermentation 9221B, SMEWW 2017 23 rd ed
13.7 Bottled water (Mineral water)	Heterotrophic Plate Count cfu/mL	Pour Plate Method 9215B, SMEWW 2017 23 rd ed
	Coliforms, MPN/100 mL	Multiple Tube Fermentation 9221B, SMEWW 2017 23 rd ed
13.8 Chilled young coconut	Aerobic Plate Count, cfu/mL	Pour Plate Method BAM Online Chapter 3
	Yeast & Mold Count, cfu/mL	Spread Plate Technique/ BAM Online Chapter 18
	Coliforms. cfu/mL	Pour Plate Method/ BAM Online Chapter 4
14. FOOD FOR INFANTS AND YOUNG CHILDREN		
14.1 Powdered Infant Formula with or without added lactic acid producing cultures (intended for 0 to 6 months old)	<i>Salmonella</i>	Reveal 2 for Salmonella
14.2 Infant Formula liquid (UHT/sterilized)	Commercial sterility	BAM Online Chapter 21A
14.3 Baby foods in hermetically sealed containers	Commercial sterility	BAM Online Chapter 21A
14.4 Dried and Instant products requiring reconstitution	Coliforms, MPN/g	MPN Method BAM Online Chapter 4
	Aerobic Plate Count, cfu/mL	Pour Plate Method/ BAM Online, Chapter 3
	<i>Salmonella</i>	Reveal 2 for Salmonella
14.5 Dried products requiring reconstitution and boiling before consumption	Coliforms, cfu/g	Pour Plate Method/ BAM Online Chapter 4
	<i>Salmonella</i>	Reveal 2 for Salmonella
	Aerobic Plate Count, cfu/mL	Pour Plate Method/ BAM Online, Chapter 3
14.6 Cereal based foods for infants		
14.7 Ready-to-Use Therapeutic Foods (RUTF) and Ready-to Use	<i>Salmonella</i>	Reveal 2 for Salmonella

Products	Specific Test	Method/Reference Standard
Supplementary Foods (RUSF), 6-59 months of age		

Legend to Reference Standards:

BAM – Bacteriological Analytical Manual

SMEWW – Standard Methods for the Examination of Water and Wastewater

****Nothing follows****